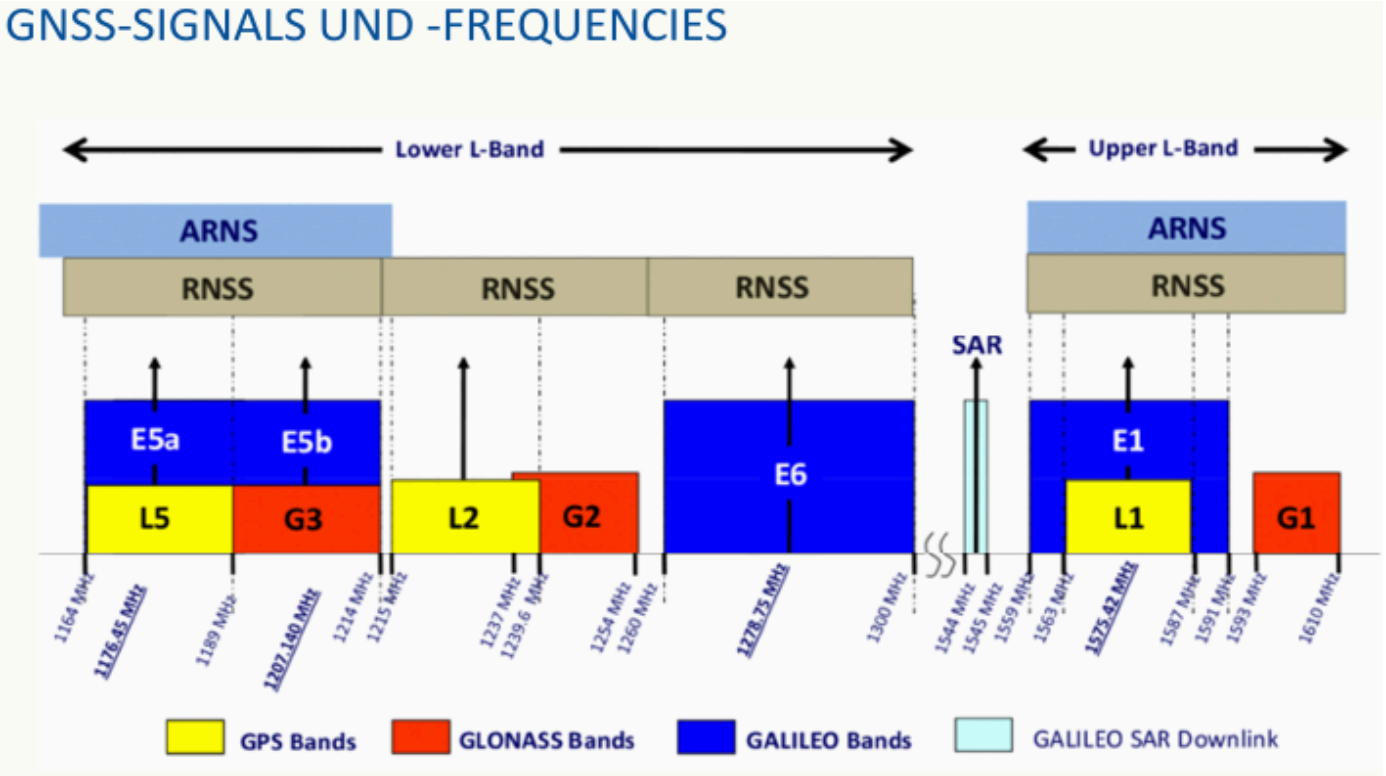
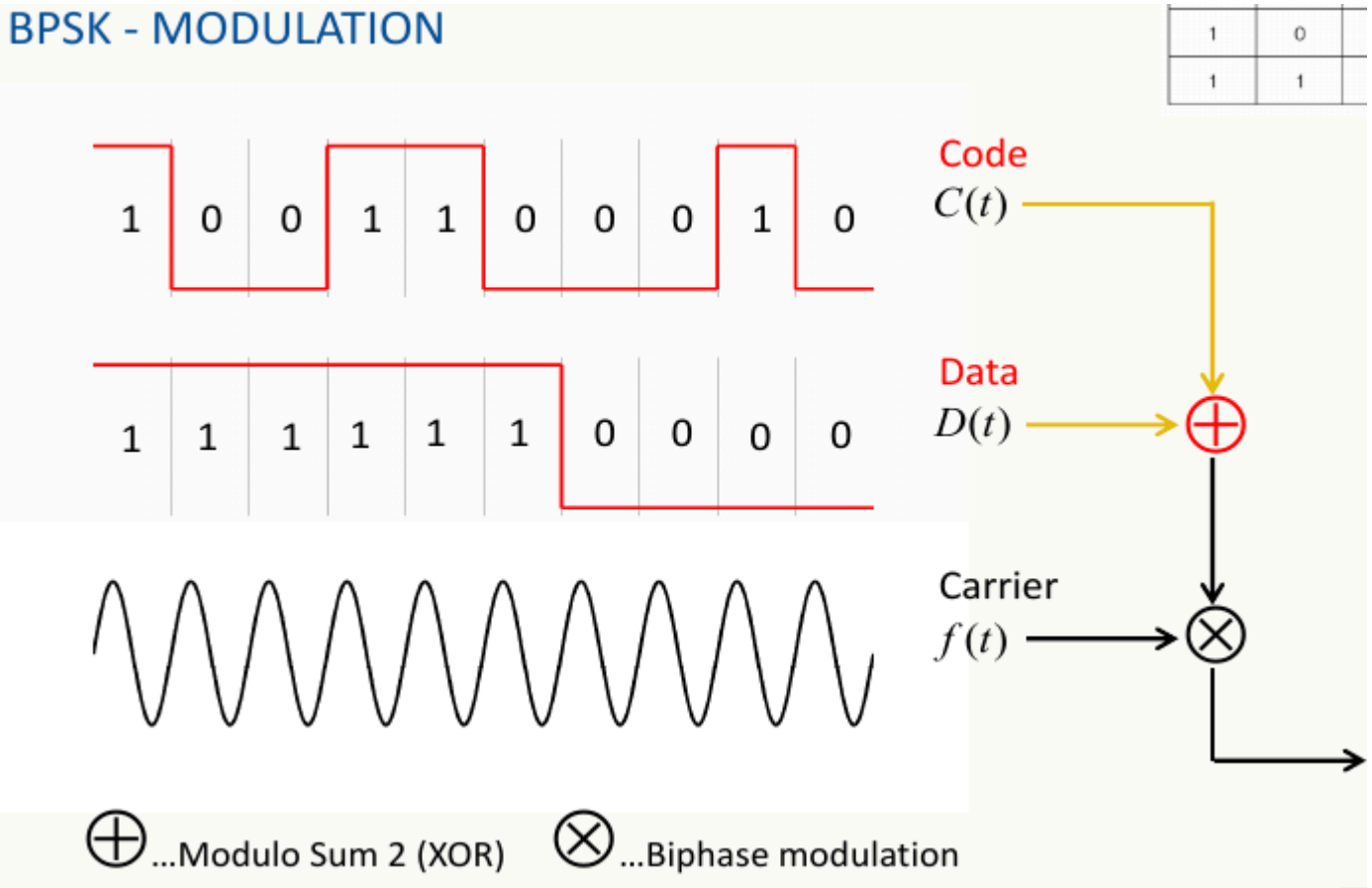
Lecture 3: Signals, Receivers and Observations

- In UHF (Ultra High frequency) band: **300MHz - 3GHz**
- GPS, GLONASS, BeiDou, Galileo, QZSS, IRNSS/NavIC share some frequency bands

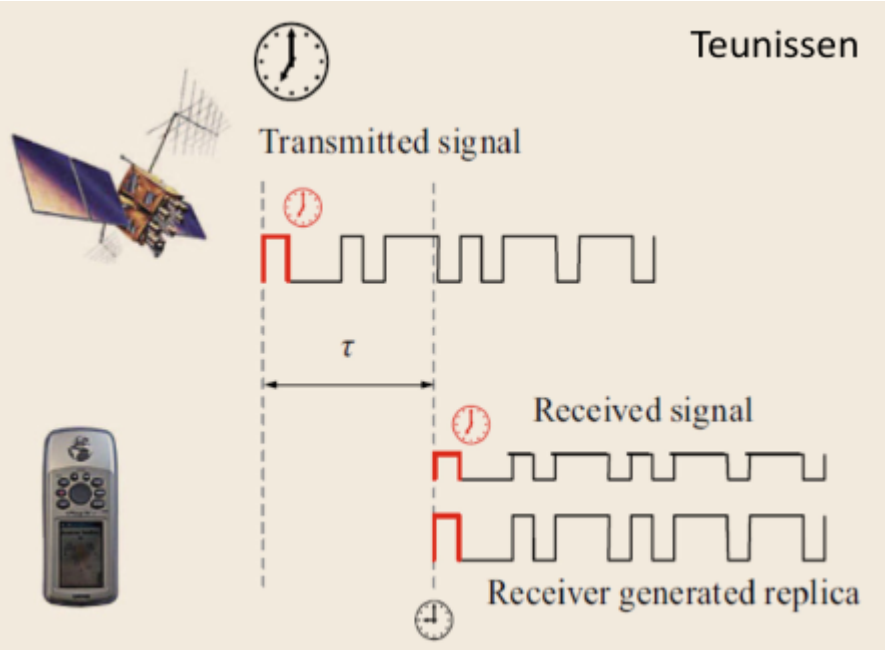


- GPS uses Phase Modulation with BPSK and Binary Offset Carrier (BOC)



Code Based Ranging (Synchronised Clocks)

1. Satellite transmits signal (**code**) at t
2. Receiver Internal Replica
3. Constant Shifting and Comparison between both signals (**correlation**) until fully overlapping (**Lock**)
4. **Transit Time** τ between t and lock time (Range = $c \cdot \tau$) (~dm-m)
5. Receiver keeps track of the shift (**Tracking**)



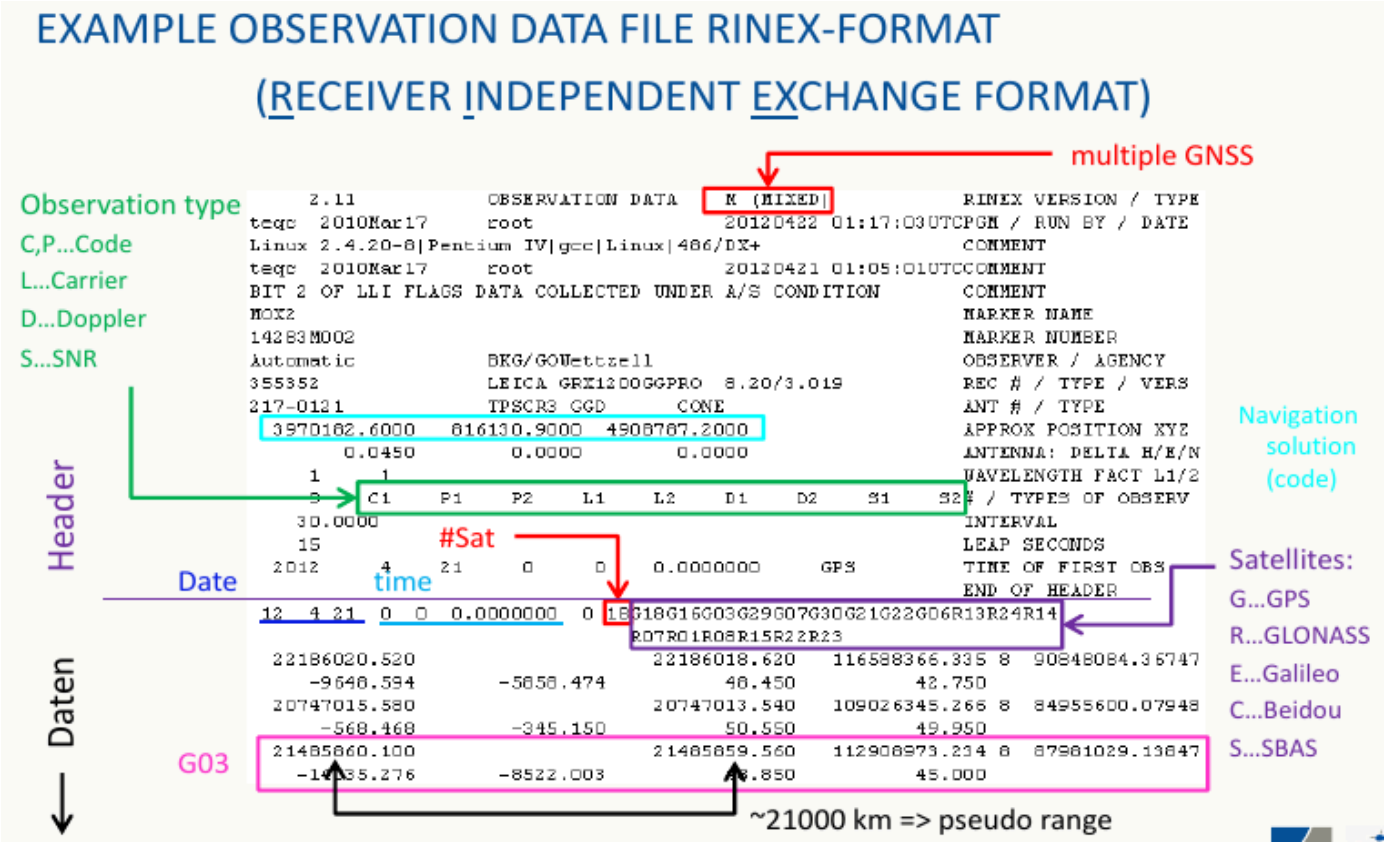
- Every satellite has its own code
- Shorter chip length -> higher resolution of the autocorrelation function
- low autocorrelation for all dt $\neq$ 0 $\rightarrow$ sharper peaks
- Low crosscorrelation with all other codes at all dt
  - -> less interference with other satellites
- PRN codes are carefully designed (so-called gold codes)

GPS Code Types

C/A Code (Coarse /Aquisition)	P(Y) Code (Precision), Encrypted
one code per satellite	
Transmitted on L1 frequency	Transmitted on L1 and L2 frequencies
Chip length = ~300m	30m
Code repeats after 1ms	after 7 days
range accuracy ~1-10m	~0.1-1m

Receiver

- **radio frequency front-end**
  - signal reception, preprocessing, filtering and gain
- **digital signal processor**
  - acquisition and tracking, extraction of observables
- **navigation processor**
  - calculation of position and velocity

GNSS - Observation data (per satellite)

- Code Measurements
  - Signal time of flight including clock offsets based on code correlation
  - Multiple codes on multiple frequencies
- Carrier Phase Measurements
  - Clock offset and unknown number of full wavelength (ambiguities) based on carrier wave correlation
- Doppler Shift
- SNR
  - $\frac{P_{signal}}{P_{noise}}$
  - $SNR(DB) = 10log_{10}(\frac{P_{signal}}{P_{noise}}) = P_{signal}(dB) - P_{noise}(dB)$
  - ~ 35dB-Hz to 45dB-Hz